Assignment 1 - Git & GitHub Fundamentals

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GitHub username: account handle not yet confirmed. Display name supplied: Hafsa mhf.

> Completion status: the local workflow is complete. GitHub publication, the online README edit, fetch/pull evidence, and URL submission have not yet been verified.

1. Repository Setup

The supplied STARTER files were initialized with git init -b main. The .gitignore rules below exclude local.env and ordinary log files while keeping important.log available to Git.

```
local.env
*.log
!important.log
```

Troubleshooting: *.log initially hid important.log. Adding !important.log after that pattern solved the issue. Actual before/after outputs are included in the evidence folder; important.log is tracked, while local.env, app.log and debug.log are ignored.

2. README & Commit History

README.md includes the supplied identity, a project description, technologies, a Bash code block, a documentation link, and assets/project-banner.svg. The exact GitHub handle must be confirmed and inserted before submission.

Five content commits cover setup, an HTML overview change, CSS styling, the JavaScript message, and the About feature. A merge commit integrates the feature; a seventh commit adds the report and recorded evidence.

Commit	Change
86c89fe	Initialize starter project with README and ignore rules
92e9dfa	Clarify the project overview
d2ad1f8	Improve section spacing and borders
2e57c24	Update the project status message
813ba1c	Add the About feature on its own branch
a46ee83	Merge the About feature into main
1489f7e	Add the AI-disclosed local report and workflow evidence

3. Selective Staging

index.html and style.css were modified together. Only index.html was staged with git add index.html and committed first. style.css remained modified and was committed separately afterward.

```
Selective staging: only index.html is staged; the CSS change remains unstaged. Both diff types are
 shown.
Actual output captured in the AI-assisted execution environment.
UTC: 2026-10-12T12:02:58+00:00

$ git status --short
M  index.html
 M style.css
```

Figure 1. Selective staging: index.html is staged while style.css remains unstaged.

In short status, the first column describes the index and the second the working tree: index.html is staged; style.css is unstaged.

```
After the HTML-only commit, style.css remains modified and uncommitted.
Actual output captured in the AI-assisted execution environment.
UTC: 2026-10-12T12:02:58+00:00

$ git status --short
 M style.css

$ git diff -- style.css
diff --git a/style.css b/style.css
index b93e076..fae9c24 100644
--- a/style.css
+++ b/style.css
@@ -19,3 +19,11 @@ section {
   padding: 20px;
   border: 1px solid #e3e3e3;
 }
+
+/* Make individual sections easier to distinguish. */
+main section {
+  padding: 1rem;
+  margin-bottom: 1rem;
+  border: 1px solid #d5dce5;
+  border-radius: 6px;
+}
```

Figure 2. After the HTML-only commit, style.css remains modified. The unstaged diff shows its changes.

git diff shows unstaged changes; git diff --cached shows staged changes. Both actual diffs are preserved in evidence/04_selective_staging.txt; the remaining CSS diff is in evidence/05_selective_commit.txt.

4. Branch & Merge

git switch -c feature-about created the feature branch. The About section was added and committed there. After git switch main, git merge --no-ff feature-about integrated the work with an explicit merge commit. The feature was absent from main before merging and present afterward.

```
The About feature has been merged into main with an explicit merge commit.
Actual output captured in the AI-assisted execution environment.
UTC: 2026-10-12T12:02:58+00:00

$ git branch --list
  feature-about
* main

$ git log --graph --oneline --decorate --all
*   a46ee83 (HEAD -> main) Merge the About feature into main
|\
| * 813ba1c (feature-about) Add the About feature on its own branch
|/
* 2e57c24 Update the project status message
* d2ad1f8 Improve section spacing and borders
* 92e9dfa Clarify the project overview
* 86c89fe Initialize starter project with README and ignore rules
```

Figure 3. The feature branch and explicit merge are visible in the recorded Git history.

The merge capture was recorded before the report/evidence commit. Figures are rendered images of actual command output from the AI-assisted environment.

Ignore-rule verification

```
Troubleshooting resolution: the exception keeps important.log visible while normal logs and local.env remain
ignored.
Actual output captured in the AI-assisted execution environment.
UTC: 2026-10-12T12:02:58+00:00

$ git status --short --ignored
?? .gitignore
?? app.js
?? assets/
?? important.log
?? index.html
?? notes.txt
?? style.css
!! app.log
!! debug.log
!! local.env

$ git check-ignore -v local.env app.log debug.log
.gitignore:1:local.env  local.env
.gitignore:2:*.log      app.log
.gitignore:2:*.log      debug.log
```

Figure 4. The ignore exception keeps important.log visible; ordinary logs and local.env are ignored. This capture predates the initial commit.

5. Remote Changes

Command	Purpose
commit	Save the staged changes in the local history.
push	Publish local commits to a remote repository.
fetch	Download remote commits and update remote-tracking references without modifying the working files.
pull	Fetch and integrate remote changes into the current branch.

Required account work: create the empty GitHub repository with the confirmed handle, push main, edit README directly on GitHub, fetch first, and capture main behind origin/main before pulling. Then pull and verify that the online edit is present locally. No account-specific output is included because these steps have not yet been verified.

Finish_GitHub.sh guides these operations and records the actual evidence. The report must be updated with that evidence, committed inside the repository, and pushed before its URL is submitted in the shared Excel sheet.

6. Fair Use of AI

ChatGPT was used to interpret the assignment, prepare the README and source edits, execute the local Git workflow, and compose the report after my laptop became unresponsive. My prompts requested step-by-step help and then an assignment package. The package uses the proposed ignore rules, commit sequence, and About feature; the identity was matched to the details I supplied, and the report uses black-and-white text with the original dark command captures inserted as figures. The assistant checked actual Git status, staged and unstaged diffs, commit history, ignore rules, and branch/merge results, and verified that a Git bundle restored the seven commits. These checks occurred in the assistant's environment. I still need to review the commands and verify the GitHub operations before submission. Local commits use my supplied name and the temporary email git-assignment@localhost.

7. Sources & Reflection

Sources: Assignment 1 and Report Template - group A.pdf; Git presentation Session 1 - group A.pdf; Assignment1_Starter - group A.zip. AI assistance: ChatGPT, as described above.

The concept that requires the most attention in this workflow is fetch versus pull: fetch updates remote information, while pull also integrates changes. If Git behaves unexpectedly, I would first inspect git status, the active branch, and git remote -v before trying to change the repository.

Evidence is preserved as verbatim command-output text and rendered capture images. The local workflow is documented accurately; unverified account actions are not presented as completed.